

Gordon Euhyun Moon (문의현)

CONTACT INFORMATION	<i>E-mail:</i> ehmoon@sogang.ac.kr <i>Webpage:</i> https://gordonmoon.github.io <i>Address:</i> Sogang University, Adam Schall Hall, 35, Baekbeom-ro, Mapo-gu, Seoul, South Korea <i>Phone:</i> +82-2-705-8487	
RESEARCH INTERESTS	Deep Learning, High Performance Computing, Deep Learning Accelerators	
EDUCATION	The Ohio State University , Columbus, OH USA Ph.D. in Computer Science and Engineering, 2019 Adviser: Ponnuswamy Sadayappan Indiana University , Bloomington, IN USA M.S. Computer Science, 2013 Yonsei University , Seoul, South Korea B.S. in Computer Science & Industrial Engineering, 2011	
PROFESSIONAL APPOINTMENTS	Sogang University , Seoul, South Korea <i>Associate Professor - Department of Computer Science and Engineering</i> <i>Assistant Professor - Department of Computer Science and Engineering</i> <i>Adjunct Professor - Department of Economics</i> <i>Adjunct Professor - Department of Artificial Intelligence</i>	2025–present 2022–2025 2024–present 2022–present
	Korea Aerospace University , Goyang, South Korea <i>Assistant Professor - Department of Software</i>	2021–2022
	Sandia National Laboratories , Albuquerque, NM USA <i>Postdoctoral Researcher - Center for Computing Research</i>	2019–2021
RESEARCH GRANTS	National Research Foundation of Korea (NRF) , PI, ₩850,000,000, Optimizing Distributed Deep Learning and Federated Learning for Accelerating Large-Scale Deep Learning Models, 2024–2027 National Research Foundation of Korea (NRF) , Co-PI, ₩1,500,000,000, Developing a High-Performance Computing-Data Platform for Accelerating Large-Scale Machine Learning (PI: Young-jae Kim), 2024–2027 Institute for Information & Communications Technology Planning & Evaluation (IITP) , Co-PI, Information Technology Research Center (ITRC), Developing an Open Platform for Web 3.0 (PI: Sooyong Park), 2023–2030 New Faculty Research Fund , PI, Sogang University, 2022–2025 National Research Foundation of Korea (NRF) , PI, Architecture-aware Parallel Algorithms for Accelerating Training of Deep Neural Networks, 2021–2024	
AWARDS	University Outstanding Teaching Award , Sogang University, 2023–2024 Best Paper Award , KIISE Korea Software Congress (KSC), 2023 Graduate Fellowship , Indiana University, 2011–2012	

TEACHING

CSE 3030: Introduction to Computer Systems (Spring 2026)
CSE 6473: Advanced GPU Programming (Fall 2025)
CSE 4200: Recommender Systems (Fall 2025)
GITF 310: Parallel Computing (Spring 2025)
CSE 3030: Introduction to Computer Systems (Spring 2025)
COR 1010: AI Programming (Spring 2025)
CSE 6473: Advanced GPU Programming (Fall 2024)
CSE 4200: Recommender Systems (Fall 2024)
CSE 3030: Introduction to Computer Systems (Spring 2024)
COR 1010: AI Programming (Spring 2024)
CSE 6473: Advanced GPU Programming (Fall 2023)
CSE 4070: Operating Systems (Fall 2023)
CSE 3030: Introduction to Computer Systems (Spring 2023)
COR 1010: AI Programming (Spring 2023)
CSE 2035: Computer Programming II (Fall 2022)
CSE 4070: Operating Systems (Fall 2022)
SW 4432: Recommender Systems (Spring 2022)
SW 4206: AI Programming (Spring 2022)
SW 4429: Parallel Computing (Fall 2021)
SW 4205: System Programming (Fall 2021)
SW 4206: AI Programming (Spring 2021)

SELECTED PUBLICATIONS

D-HAT: Dynamic Hypergraph Representation Learning with Attention-Based Multi-Level Hypergraph Sampling
Ah-hyun Lee and **Gordon E. Moon**
Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM 2025)

EQUILIBRIA: Co-Optimizing Energy and Latency in Online ML-based Stream Processing Systems
Sejeong Oh, Soyang Baek, **Gordon E. Moon** and Sungyong Park
Proceedings of the 25th IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid 2025)

ML-based Dynamic Operator-Level Query Mapping for Stream Processing Systems in Heterogeneous Computing Environments
Sejeong Oh, **Gordon E. Moon** and Sungyong Park
Proceedings of the 26th IEEE International Conference on Cluster Computing (CLUSTER 2024)

Accelerated Block-Sparsity-Aware Matrix Reordering for Leveraging Tensor Cores in Sparse Matrix–Multivector Multiplication
Eunji Lee, Yoonsang Han and **Gordon E. Moon**
Proceedings of the 30th International European Conference on Parallel and Distributed Computing (Euro-Par 2024)

Layer-Wise Sparse Training of Transformer via Convolutional Flood Filling
Bokyeong Yoon, Yoonsang Han and **Gordon E. Moon**
Proceedings of the 28th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2024)

Chronica: A Data-Imbalance-Aware Scheduler for Distributed Deep Learning
Sanha Maeng, **Gordon E. Moon** and Sungyong Park
Proceedings of the 23rd IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGrid 2023)

Parallel Training of GRU Networks with a Multi-Grid Solver for Long Sequences

Gordon E. Moon and Eric C. Cyr

Proceedings of the 10th International Conference on Learning Representations (ICLR 2022)

Evaluating Spatial Accelerator Architectures with Tiled Matrix-Matrix Multiplication

Gordon E. Moon, Hyoukjun Kwon, Geonhwa Jeong, Prasanth Chatarasi, Sivasankaran Rajamanickam and Tushar Krishna

IEEE Transactions on Parallel and Distributed Systems (TPDS 2022)

Extending Sparse Tensor Accelerators to Support Multiple Compression Formats

Eric Qin, Geonhwa Jeong, William Won, Sheng-Chun Kao, Hyoukjun Kwon, Sudarshan Srinivasan, Dipankar Das, **Gordon E. Moon**, Sivasankaran Rajamanickam and Tushar Krishna

Proceedings of the 35th IEEE International Parallel & Distributed Processing Symposium (IPDPS 2021)

ALO-NMF: Accelerated Locality-Optimized Non-negative Matrix Factorization

Gordon E. Moon, J. Austin Ellis, Aravind Sukumaran-Rajam, Srinivasan Parthasarathy and P. Sadayappan

Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2020)

PROFESSIONAL
SERVICE

Head, Global AI Education Center, Sogang University, 2024–present

Chair, Major in Software Engineering & Computer Systems, Graduate School of AI-SW, Sogang University, 2023–present

Program Committee Member, Korea Information Processing Society (ASK 2025)

Program Committee Member, The 40th IEEE International Conference on Computer Design (ICCD 2022)

Program Committee Member, Tenth International Workshop on Accelerating Analytics and Data Management Systems Using Modern Processor and Storage Architectures (ADMS 2019)

Korean Military Service, Sixth Army Corps Headquarters, 2007–2009

INVITED TALKS

HPC Society Conference, KIISE, July 2024

HPC Society Conference, KIISE, July 2022

Korea Computer Congress (KCC), KIISE, June 2022

Computer System Society Conference, The Korean Institute of Information Scientists and Engineers (KIISE), February 2022

NVIDIA GTC, November 2021

AMS Fall Western Virtual Sectional Meeting, October 2021

SIAM Conference on Computational Science and Engineering (CSE), March 2021

Sandia National Laboratories, August 2019

Oak Ridge National Laboratory, August 2019

AMD Research, July 2019

PH.D. STUDENTS

Wonho Lee

Yoonsang Han

Ah-Hyun Lee

M.S. STUDENTS

Donghwi Seo

Seyeon Choi

Junha Park

Sumin Choi

Yeeun Ahn

Juhyeon Lee

Bokki Min
Sangwon Bae

ALUMNI

Kisun Seo (M.S., Sogang University, 2026)
Yoonsang Han (M.S., Sogang University, 2025, Joined Sogang University for Ph.D. in Computer Science and Engineering)
Eunji Lee (M.S., Sogang University, 2025, First employment: Rebellions)
Cheolhee Kim (M.S., Sogang University, 2024)
Bokyeong Yoon (M.S., Sogang University, 2024, Joined The Ohio State University for Ph.D. in Computer Science and Engineering under the supervision of Professor Martin Kong)

MISCELLANEOUS

U.S. and Korea dual citizenship